

Intravenous Infusion Titration

Drug	Standard Concentration	Bolus (Requires prescriber order)	Initial rate (Unless ordered by prescriber)	Titration (Unless defined by prescriber)	Maximum Dose (Unless defined by prescriber)	Monitoring Effects	Adverse Effects
Aminocaproic Acid (Amicar)	5 gm/250 mL	5 gm over 1 hr	1 gm/hr for ~8 hr or until bleeding stops		36 gm/24 hrs	Fibrinogen, fibrin split products, CPK in long term therapy	Hypotension, bradycardia, rhabdomyolysis
Amiodarone (Cordarone)	Peripheral conc. 450 mg/250 mL 360 mg/200 mL Central conc. 750 mg/250 mL	150 mg over 10 mins	1 mg/min x 6 hr then 0.5 mg/min		2000 mg/24 hrs	HR, EKG, BP, signs of thyroid dysfunction and pulmonary toxicity	Hypotension, bradycardia
Angiotensin II (Giapreza)	2.5 mg/500 mL (NS only)	None	10 ng/kg/min	titrate every 5 min by increments of 15 ng/kg/min	80 ng/kg/min during the first 3 hr, thereafter, maximum dose is 40 ng/kg/min	BP	Thrombosis, Delirium, Thrombocytopenia, Fungal infections
Argatroban (Argatroban)	250 mg/250 mL	<u>PCI:</u> 350 mcg/kg over 5 min	<u>H.I.T.:</u> 2 mcg/kg/min <u>PCI:</u> 25 mcg/kg/min	<u>H.I.T.:</u> PTT at baseline and 4 hr after rate changes <u>PCI:</u> Measure ACT 5-10 minutes after any bolus dose or rate change	<u>H.I.T.:</u> 10 mcg/kg/min	Baseline PTT, monitor H&H and signs of bleeding	Bleeding at any site, decreased hemoglobin/ hematocrit, fever, and hypotension
Bumetanide (Bumex)	25 mg/100 mL	0.5 – 1 mg over 2 min	0.5 mg/hr		4 mg/hr	BP, serum creatinine, electrolytes	Hypotension, renal failure, hypokalemia, hypovolemia
Cisatracurium (Nimbex)	200 mg/100 mL	0.1-0.2 mg/kg for intubation	0.5 mcg/kg/min	Use peripheral nerve stimulator. Titrate by 0.5 mcg/kg/min every 30 min according to desired TOF	10 mcg/kg/min	HR, BP, respiratory rate, TOF	Bradycardia, hypotension, bronchospasm
Clevidipine (Cleviprex)	25 mg/50 mL		1 mg/hr	titrate by 1 mg/hr at 5 min intervals. When at goal BP, titrate by 1 mg/hr at 5 to 10 min intervals	21 mg/hr or 72 hr duration	BP, HR, heart rhythm	Hypotension, reflex tachycardia, nausea, A.fib, pyrexia
Dexmedetomidine (Precedex)	200 mcg/250 mL	1 mcg/kg over 10 min. May omit loading dose if converting from another sedative or concern for hemodynamic compromise	0.2 mcg/kg/hr	titrate by 0.1 mcg/kg/hr every 30 min to level of desired sedation (RASS score)	0.7 mcg/kg/hr	Level of sedation (RASS score), heart rate, respiration, rhythm	Hypotension, nausea, bradycardia, A.fib, hypoxia
Diltiazem (Cardizem)	125 mg/125 mL	0.25 mg/kg over 2 min. May repeat bolus at 0.35 mg/kg in 15 minutes if inadequate response	5 mg/hr	titrate rate by 5 mg/hr at 15 min intervals	15 mg/hr	BP, HR and EKG	AV block, edema, hypotension, bradycardia
Dobutamine (Dobutrex)	1000 mg/250 mL		2.5 mcg/kg/min	titrate by 2.5 mcg/kg/min at 3 min intervals	40 mcg/kg/min	BP, EKG, HR	Arrhythmias, HTN, ectopic heartbeats, angina, N/V, HA, ventricular tachycardia

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Dopamine (Intropin)	800 mg/250 mL		2 mcg/kg/min	1 mcg/kg/min at 10 min intervals	20 mcg/kg/min	EKG, HR, BP, urine output	HTN, arrhythmias hypotension, angina N/V, headache
Epinephrine (Adrenalin)	8 mg/250 mL	1 mg every 3-5 min as needed	1 mcg/min	titrate by 1 mcg/min	20 mcg/min	HR, BP, extravasation	HTN, tachycardia, arrhythmias, headache, N/V
Esmolol (Brevibloc)	Peripheral conc. 2500 mg/250 mL Central conc. 2000 mg/100 mL 5 gm/250 mL	0.5 mg/kg over 1 min	50 mcg/kg/min	titrate by 50 mcg/kg/min at 4 min intervals	300 mcg/kg/min	BP, HR, MAP, EKG, respiratory rate	Hypotension, bradycardia, nausea
Fenoldopam (Corlopam)	10 mg/250 mL		0.1 mcg/kg/min	titrate every 15 min at 0.05 mcg/kg/min increments to goal BP	1.6 mcg/kg/min	BP, HR, EKG	Hypotension, tachycardia, angina, dizziness, blurred vision
Fentanyl (Sublimaze)	1000 mcg/100 mL	<u>Premedication, Regional Anesthesia, or Severe Pain:</u> 25-100 mcg/dose over 1 min	<u>(on ventilator):</u> 25 mcg/hr	titrate by 10 mcg/hr every 10 min according to pain score	150 mcg/hr. Varies due to tolerance – limited by respiratory depression	Respiratory and cardiovascular status, BP, HR, pain score	Bradycardia, hypotension, drowsiness, sedation, increases ICP, N/V, constipation, respiratory & CNS depression
Furosemide (Lasix)	100 mg/100 mL	20-40 mg over 2 min	2 mg/hr		40 mg/hr	BP, serum creatinine, electrolytes	Hypotension, renal failure, hypokalemia, hypovolemia
Heparin	25,000 units/250 mL In D5W Concentration or diluent shall not be changed	<u>DVT/PE:</u> 80 Units/kg <u>MI/Unstable angina:</u> 60 Units/kg	<u>DVT/PE:</u> 18 Units/kg/hr <u>MI/Unstable angina:</u> 12 Units/kg/hr	TT at baseline and 6 hr after bolus or 6 hr after infusion rate changes	To achieve PTT 1.5-2.5 x control	Platelet counts, PTT, hemoglobin, hematocrit, signs of bleeding	Thrombocytopenia, skin necrosis, bleeding at any site
Isoproterenol (Isuprel)	2 mg/250 mL		2 mcg/min	titrate by 1 mcg/min at 15 min intervals	10 mcg/min	EKG, HR, RR, BP	Arrhythmia, hypotension, tachycardia, HA, anxiety, angina, dizziness, tinnitus
Ketamine/Fentanyl (Refer to Continuous Ketamine/Fentanyl Intravenous Infusion policy)	500 mg/1250 mcg in 250 mL NS	None	0.05 mg/kg/hr (based on Ketamine)	titrate by ____ at ____ min intervals Physician/provider order is required for each dose change above 10 mL/hr	0.4 mg/kg/hr (based on Ketamine)	Heart rate, BP, respiratory rate, O ₂ saturation, continuous cardiac function in patients with increased BP or cardiac decompensation	Bradycardia, hypotension, sedation, increases ICP, N/V, constipation, respiratory & CNS depression, Hallucinations, psychological disturbances
Labetalol (Trandate)	200 mg/100 mL	20 mg over 2 min	0.5 mg/min	titrate by 0.5 mg/min at 10 min intervals	3 mg/min	BP, HR, MAP, EKG, respiratory rate	Hypotension, bradycardia, nausea
Lidocaine (Xylocaine)	2 gm/500 mL	1-1.5 mg/kg over 2-3 min	1 mg/min		4 mg/min	EKG, CNS, phlebitis	Slurred speech, drowsiness, altered consciousness, seizures
Lorazepam (Ativan)	60 mg/60 mL	<u>STATUS EPILEPTICUS</u> 0.05-0.15 mg/kg (maximum 4 mg) over 2 min	<u>SEDATION:</u> <u>(on ventilator)</u> 1 mg/hr	titrate by 1 mg/hr at 30 min intervals according to desired sedation level (RASS score)	4 mg/hr	Respiratory and cardiovascular status, BP, RASS score	Hypotension, respiratory depression, oversedation, anterograde amnesia, physical dependence

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Midazolam (Versed)	100 mg/100 mL	<u>SEDATION:</u> 0.01-0.05 mg/kg, may repeat every 5-10 min as needed	<u>SEDATION:</u> <u>(on ventilator)</u> 1 mg/hr	titrate by 1mg/hr at 15 min intervals according to desired sedation level (RASS score)	20 mg/hr	Respiratory and cardiovascular status, BP, RASS score	Hypotension, respiratory depression, oversedation, anterograde amnesia, physical dependence
Milrinone (Primacor)	20 mg/100 mL	50 mcg/kg over 10 min	0.25 mcg/kg/min		1 mcg/kg/min	BP, hemodynamic responses (PCWP, RAP, CI)	Ventricular and supraventricular arrhythmias, hypotension, headache
Nicardipine (Cardene)	<u>Peripheral conc.</u> 20 mg/200 mL 40 mg/200 mL <u>Central conc.</u> 50 mg/100 mL		<u>Hypertension:</u> 5 mg/hr	titrate by 1 mg/hr every 5 min	15 mg/hr	BP, HR, peripheral edema	Flushing, palpitation, tachycardia, peripheral edema, angina, HA, dizziness
Nitroglycerin	50 mg/250 mL		5 mcg/min	titrate by 5 mcg/min every 3-5 min	200 mcg/min	BP, HR	Hypotension, HA, dizziness N/V, syncope
Nitroprusside (Nipride)	50 mg /250mL		0.5 mcg/kg/min	titrate by 0.5 mcg/kg/min at 3-5 min intervals	2 mcg/kg/min (higher doses for up to 10 minutes may be ordered)	BP, HR, Cyanide & thiocyanate toxicity, acid- base status	Hypotension, palpitation, HA, psychosis, N/V
Norepinephrine (Levophed)	8 mg/250 mL		0.5 mcg/min	titrate by 0.5 mcg/min every 3-5 min	30 mcg/min	BP, HR	Extravasation (leading to necrosis) HA, N/V, tachycardia, HTN, arrhythmias, urinary retention
Pentobarbital (Nembutal)	2500 mg/500 mL	10-15 mg/kg over 1 hr	0.5 mg/kg/hr	titrate by 0.5mg/kg/hr at 1 hr intervals according to desired sedation level (RASS score)	3 mg/kg/hr (monitor levels)	BP, EEG, daily sedation levels (RASS score)	Respiratory depression, hypotension, cardiac arrhythmias
Phenylephrine (Neo-Synephrine)	40 mg/250 mL		50 mcg/min	titrate by 25 mcg/min at 5 min intervals	200 mcg/min	BP	Hypertension, impaired myocardial perfusion
Procainamide (Pronestyl)	2 gm/250 mL	1 gm over 1 hr	1 mg/min		6 mg/min (monitor levels)	EKG, CBC, BP, platelet count, cardiac, liver and renal function	Rash, diarrhea, N/V, taste disorder, hypotension, angioedema
Propofol (Diprivan)	1000 mg/100 mL		<u>(on ventilator)</u> 5 mcg/kg/min	titrate by 5 mcg/kg/min every 5-10 min according to desired sedation level (RASS score)	100 mcg/kg/min	BP, serum triglycerides, daily sedation levels (RASS score)	Hypotension, apnea, injection site pain
Vasopressin (Pitressin)	40 units/100 mL	<u>Pulseless VT/VF:</u> 40 units IV dose	<u>Shock:</u> 0.02 units/min <u>GI Hemorrhage:</u> 0.2 units/min	<u>Shock:</u> titrate by 0.01 units/min <u>GI Hemorrhage:</u> titrate by 0.2 unit/min every 1 hr	<u>Shock</u> 0.04 unit/min <u>GI Hemorrhage</u> 0.8 unit/min	BP, serum and urine sodium, fluid input and output, urine specific gravity, urine and serum osmolality	Hypertension, ventricular arrhythmias, angina, tremor, diaphoresis
Vecuronium (Norcuron)	100 mg/100 mL	0.05-0.1 mg/kg bolus	0.7 mcg/kg/min	titrate by 0.1 mcg/kg/min at 60 min intervals according to desired TOF	1 mcg/kg/min	BP, HR, TOF	Bradycardia, circulatory collapse, edema, flushing, hypotension, itching, rash